MATH 55 - HW 1 - Jie Wen

Problems from Hutchings notes on Introduction to Mathematical Arguments

1. Proofs of Statements about Even and Odd Integers

(a) For every integer x, if x is even, then for every integer y, xy is even.

Proof:

Let x be any integer and assume that x is even. By the definition of an even integer, there exists an integer k such that $x = 2k$.
Now, let y be any integer.
Substituting the expression for x:
$xy = (2k)y = 2(ky)$
Since k and y are integers, their product, ky, is also an integer. Denote this integer by m: $m = ky$.
Therefore, $xy = 2m$.
According to the definition of an even integer, since xy is 2 times an integer m, xy is even.

Negation of (a):

The original statement is: $(\forall x \in \mathbb{Z})(x \text{ is even} \Rightarrow (\forall y \in \mathbb{Z})(xy \text{ is even}))$.
The negation is: There exists an integer x such that x is even and there exists an integer y such that xy is not even (i.e., is odd).
In symbolic notation: $(\exists x \in \mathbb{Z})(x \text{ is even} \wedge (\exists y \in \mathbb{Z})(xy \text{ is odd}))$.

(b) For every integer x and for every integer y, if x is odd and y is odd then $x + y$ is even.

Proof:

Let x and y be any integers. Assume that both x and y are odd.
By the definition of an odd integer, there exists an integer k_1 such that $x = 2k_1 + 1$.
Similarly, there exists an integer k_2 such that $y = 2k_2 + 1$.
Now, consider the sum $x + y$:
$x + y = (2k_1 + 1) + (2k_2 + 1) = 2k_1 + 2k_2 + 2 = 2(k_1 + k_2 + 1)$
Since k_1 and k_2 are integers, the expression $k_1 + k_2 + 1$ is also an integer. Denote this integer by n: $n = k_1 + k_2 + 1$.
Therefore, $x + y = 2n$.
According to the definition of an even integer, since $x + y$ is 2 times an integer n, $x + y$ is even.

Negation of (a):

The original statement is: $(\forall x \in \mathbb{Z})(\forall y \in \mathbb{Z})((x \text{ is odd} \wedge y \text{ is odd}) \Rightarrow x + y \text{ is even})$.
The negation is: There exist integers x and y such that x is odd and y is odd, but their sum $x + y$ is not even (i.e., is odd).
In symbolic notation: $(\exists x \in \mathbb{Z})(\exists y \in \mathbb{Z})(x \text{ is odd} \wedge y \text{ is odd} \wedge x + y \text{ is odd})$.

2. Prove that for every integer x, $x + 4$ is odd if and only if $x + 7$ is even.

Part 1: If $x + 4$ is odd, then $x + 7$ is even.

Assume that $x + 4$ is odd. By definition, there exists an integer k such that $x + 4 = 2k + 1$.
Solving for x: $x = 2k - 3$.
Then, $x + 7 = (2k - 3) + 7 = 2k + 4 = 2(k + 2)$
Since k is an integer, $k + 2$ is also an integer. Let $m = k + 2$.
Thus, $x + 7 = 2m$, which shows that $x + 7$ is even by definition.

Part 2: If $x + 7$ is even, then $x + 4$ is odd.

Assume that $x + 7$ is even. By definition, there exists an integer j such that $x + 7 = 2j$.
Solving for x: $x = 2j - 7$.
Then, $x + 4 = (2j - 7) + 4 = 2j - 3 = 2(j - 2) + 1$
Since j is an integer, $j - 2$ is also an integer. Let $n = j - 2$.
Thus, $x + 4 = 2n + 1$, which shows that $x + 4$ is odd by definition.

Since both implications have been proven, the original statement is true.

3. Figure out whether the statement we negated in §1.3 is true or false, and prove it.

The statement is: $\forall x \in \mathbb{Z}\left((\exists y \in \mathbb{Z}, x = 3y + 1) \Rightarrow (\exists y \in \mathbb{Z}, x^2 = 3y + 1)\right)$.
In words: For any integer x, if x leaves a remainder of 1 when divided by 3, then x^2 also leaves a remainder of 1 when divided by 3.

The statement is **true**.

Proof:
Let x be an arbitrary integer and assume that there exists an integer y such that $x = 3y + 1$.
$x^2 = (3y + 1)^2 = 9y^2 + 6y + 1 = 3(3y^2 + 2y) + 1$
Let $z = 3y^2 + 2y$. Since y is an integer, y^2 is an integer. Consequently, $3y^2$ and $2y$ are integers, and their sum, z, is also an integer.
Therefore, x^2 can be written in the form $3z + 1$ for some integer z. This proves the original statement.

4. Prove that for every integer x, if x is odd then there exists an integer y such that $x^2 = 8y + 1$.

Let x be any integer and assume that x is odd.
By the definition of an odd integer, there exists an integer k such that $x = 2k + 1$.
$x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$

Consider the product $k(k + 1)$: this is the product of two consecutive integers.

- If k is even:
  - $k = 2m$ for some integer m. The product is $2m(2m + 1)$, which is a multiple of 2.
- If k is odd
  - $k + 1$ is even. So, $k + 1 = 2n$ for some integer n. The product is $k(2n)$, which is a multiple of 2.

In either case, the product $k(k + 1)$ is always an even integer. Therefore, we can write $k(k + 1) = 2y$ for some integer y.
Substituting this back into our expression for x^2: $x^2 = 4(2y) + 1 = 8y + 1$
So if x is an odd integer, then there exists an integer y such that $x^2 = 8y + 1$.

Problems from the Rosen textbook

Section 1.1: Propositional Logic - Problem 17

(c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

- "If berries are ripe along the trail..." translates to $r \rightarrow \ldots$
- "...hiking is safe..." is q.
- "...if and only if..." is the biconditional connective $\leftrightarrow$.
- "...grizzly bears have not been seen in the area" is the negation of p, which is $\neg p$.
- Combining the second part, "hiking is safe if and only if grizzly bears have not been seen in the area," gives us $(q \leftrightarrow \neg p)$.

Putting it all together, the full proposition is: $r \rightarrow (q \leftrightarrow \neg p)$

(f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

- "Hiking is not safe on the trail..." is $\neg q$.
- The word "whenever" functions like "if" in logic. So, we can read this as "If grizzly bears have been seen... then hiking is not safe." This gives us a structure of $(\ldots) \rightarrow \neg q$.
- The condition is "grizzly bears have been seen in the area *and* berries are ripe along the trail," which is $(p \wedge r)$.

So, the complete proposition is: $(p \wedge r) \rightarrow \neg q$

Are these two propositions logically equivalent? Why or why not?

#	p	q	r	¬p	¬q	q ↔ ¬p	r → (q ↔ ¬p)	p ∧ r	(p ∧ r) → ¬q
1	T	T	T	F	F	F	**F**	T	**F**
2	T	T	F	F	F	F	**T**	F	**T**
3	T	F	T	F	T	T	**T**	T	**T**
4	T	F	F	F	T	T	**T**	F	**T**
5	F	T	T	T	F	T	**T**	F	**T**
6	F	T	F	T	F	T	**T**	F	**T**
7	F	F	T	T	T	F	**F**	F	**T**
8	F	F	F	T	T	F	**T**	F	**T**

In the 7th row (when p is false, q is false, and r is true), the proposition from (c) is **False**, while the proposition from (f) is **True**.

Therefore, **the two propositions are not logically equivalent**.

To prove it **without** using the truth table: berries are ripe ($r = T$), but hiking is not safe ($q = F$), and no grizzly bears have been seen ($\neg p = T$).

- Statement (c) becomes: "If berries are ripe, then (hiking is not safe if and only if no bears have been seen)." This evaluates to $T \to (F \leftrightarrow T)$, which is $T \to F$, which is **False**.
- Statement (f) becomes: "If bears have been seen and berries are ripe, then hiking is not safe." This evaluates to $(F \wedge T) \to T$, which is $F \to T$, which is **True**.

Section 1.3: Propositional Equivalences, Problems 12 and 14

Problem 12: Show that each of these conditional statements is a tautology by using truth tables

(A) $[\neg P \wedge (P \vee Q)] \to Q$

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \to q$
T	T	F	T	F	**T**
T	F	F	T	F	**T**
F	T	T	T	T	**T**
F	F	T	F	F	**T**

The final column is true in all rows, so it is a tautology.

(B) $[(P \to Q) \wedge (Q \to R)] \to (P \to R)$

p	q	r	$p \to q$	$q \to r$	$(p \to q) \wedge (q \to r)$	$p \to r$	$[(p \to q) \wedge (q \to r)] \to (p \to r)$
T	T	T	T	T	T	T	**T**
T	T	F	T	F	F	F	**T**
T	F	T	F	T	F	T	**T**
T	F	F	F	T	F	F	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	F	F	T	**T**
F	F	T	T	T	T	T	**T**
F	F	F	T	T	T	T	**T**

The final column is true in all rows, so it is a tautology.

(C) $[P \wedge (P \to Q)] \to Q$

p	q	$p \to q$	$p \wedge (p \to q)$	$[p \wedge (p \to q)] \to q$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	F	**T**

The final column is true in all rows, so it is a tautology.

(D) $[(P \vee Q) \wedge (P \to R) \wedge (Q \to R)] \to R$

p	q	r	$p \vee q$	$p \to r$	$q \to r$	$(p \vee q) \wedge (p \to r) \wedge (q \to r)$	$[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$
T	T	T	T	T	T	T	**T**
T	T	F	T	F	F	F	**T**
T	F	T	T	T	T	T	**T**
T	F	F	T	F	T	F	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	T	F	F	**T**
F	F	T	F	T	T	F	**T**
F	F	F	F	T	T	F	**T**

The final column is true in all rows, so it is a tautology.

Problem 14: Show that the conditional statements in 12(b) and 12(d) are tautologies using the fact that a conditional statement is false exactly when the hypothesis is true and the conclusion is false (Do not use truth tables)

For 12(b): Assume the hypothesis $(p \to q) \wedge (q \to r)$ is true. To show $p \to r$, assume p is true (otherwise $p \to r$ is vacuously true). Then $p \to q$ implies q is true, and $q \to r$ implies r is true. Thus, the conclusion holds, so the overall conditional is true in all cases.

For 12(d): Assume the hypothesis $(p \vee q) \wedge (p \to r) \wedge (q \to r)$ is true. Since $p \vee q$ is true, either p is true or q is true (or both). If p is true, then $p \to r$ implies r is true. If q is true, then $q \to r$ implies r is true. In either case, r is true. Thus, the conclusion holds, so the overall conditional is true in all cases.

How each tautology could be used in a logical argument

For 12(a): If we know that at least one of two statements is true, and we can rule out the first, then we can conclude that the second is true.

For 12(b): If we know that one statement implies a second, and the second implies a third, then we can conclude that the first implies the third (transitivity of implications, often used to chain steps in proofs).

For 12(c): If we know a statement is true, and it implies a second statement, then we can conclude that the second is true (modus ponens, a fundamental rule of inference).

For 12(d): If we know that at least one of two statements holds, and we also know that each of them implies a third statement, then we can conclude that the third statement holds (proof by cases, where the cases cover the disjunction).

Section 1.4: Predicates and Quantifiers

46. Determine whether $\forall x(P(x) \leftrightarrow Q(x))$ and $\forall x P(x) \leftrightarrow \forall x Q(x)$ are logically equivalent.

- These two statements are **not logically equivalent**.
- Consider the domain of **all integers**.
  - Let $P(x)$ be "x is an even number."
  - Let $Q(x)$ be "x is an odd number."
- Now evaluate the two statements:
  1. $\forall x(P(x) \leftrightarrow Q(x))$
     - This statement reads: "For every integer x, x is even if and only if x is odd." This is clearly **False**. No integer is both even and odd, and every integer is one or the other.
  2. $\forall x P(x) \leftrightarrow \forall x Q(x)$
     - This statement is a biconditional between two separate quantified statements.
       - $\forall x P(x)$ reads "All integers are even." This is **False**.
       - $\forall x Q(x)$ reads "All integers are odd." This is also **False**.
       - The full statement becomes $F \leftrightarrow F$, which is **True**.
- Since there exists an interpretation where the first statement is False and the second is True, they cannot be logically equivalent.
- **Justification:** The first statement makes a very strong claim: that for *every single element* x, the properties $P(x)$ and $Q(x)$ must be in lockstep (either both true or both false). The second statement is much weaker: it only claims that the truth value of "everything has property P" is the same as the truth value of "everything has property Q".

54. Domain is all integers. What are the truth values of these statements?

- (a) $\exists! x(x > 1)$: **False**, because there are infinitely many integers greater than 1 (e.g., 2, 3, 4, ...), not exactly one.
- (c) $\exists! x(x + 3 = 2x)$: **True**, because solving gives $3 = x$, so $x = 3$ is the unique integer solution.
- (d) $\exists! x(x = x + 1)$: **False**, because this simplifies to $0 = 1$, which has no integer solutions (zero solutions, not exactly one).

Section 1.5: Nested Quantifiers

24. Translate into an English statement: $\forall x \forall y((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$

For all real numbers x and y, both x and y are nonzero if and only if their product xy is nonzero.

40. Find a counterexample, if possible, to $\forall x \exists y(y^2 - x < 100)$ where the domain is all integers.

Consider $x = -100$ (an integer). Then the inequality becomes $y^2 < 0$. But for any integer y, $y^2 \geq 0$, so no such y exists. Thus, the universal quantification fails for this x.